

Smart IoT Attendance System Ecosystem

AN INTEGRATED RFID-BASED ACADEMIC MANAGEMENT SOLUTION

6 October Technological University

Faculty of Industry and Energy Technology | Information Technology Program

GRADUATION PROJECT 2025-2026

Contents

01	Introduction Background, Problem, Objectives & Scope	02	Literature Review Attendance Systems, RFID & IoT	03	System Analysis Requirements & Use Case Analysis	04	System Design Architecture & Database
05	Implementation Backend, Frontend, Mobile & IoT	06	Testing & Results Methodology & Performance	07	Conclusion Achievements & Future Work		



01

CHAPTER ONE

Introduction

Background, Problem Statement,
Objectives & Project Scope



Background & Problem Statement

The Challenge

Traditional attendance methods consume **5-10 minutes per lecture**, accumulating to **8-16 hours** of lost instructional time per course per semester.

Iraqi universities face unique infrastructure challenges requiring **offline-first solutions** that operate without internet connectivity.



Key Problems Identified

- **Internet Dependency** — Cloud systems fail offline
- **Manual Inefficiency** — 5-10 min lost per session
- **Proxy Attendance** — Fraudulent sign-ins
- **Data Fragmentation** — Records scattered
- **Limited Analytics** — No insights provided
- **Multi-Faculty Complexity** — Hierarchical needs

5-10

Minutes lost per lecture
with manual methods

8-16

Hours lost per course per
semester

6

Critical challenges
identified in Iraqi
universities

Project Objectives & Scope

OBJECTIVES



Offline-Capable System
No internet required for full operation



RFID-Based Identification
ESP32 + RC522 automated scanning



6-Tier RBAC Security
Role-based access control



React Web Dashboard
Administrative interface & analytics



Kotlin Android App
Mobile access for students



AI-Powered Analytics
RAG-based academic insights

PROJECT SCOPE

In Scope

- ✓ RFID-based attendance scanning
- ✓ Multi-faculty academic structure
- ✓ 6-tier role-based access control
- ✓ Web dashboard with analytics
- ✓ Android mobile application
- ✓ Gradebook with 40/60 weighting
- ✓ AI academic query system
- ✓ Raspberry Pi offline deployment

Out of Scope

- ✗ Biometric authentication
- ✗ iOS mobile application
- ✗ Cloud-based deployment
- ✗ ERP system integration
- ✗ Payment/financial features

02

CHAPTER TWO

Literature Review

Attendance Systems Evolution,
RFID Technology & IoT in Education



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Attendance Systems Evolution & RFID

EVOLUTION TIMELINE

1. Manual Systems

Roll calls & sign-in sheets

5-10 min per session

Error-prone & fraudulent

No analytics capability

2. Barcode Systems

Laser scanning of cards

Line-of-sight required

Sensitive to card damage

Limited storage capacity

3. Biometric Systems

Fingerprint / facial recognition

High security, high cost

Privacy concerns

Environmental sensitivity

4. RFID Systems (Ours)

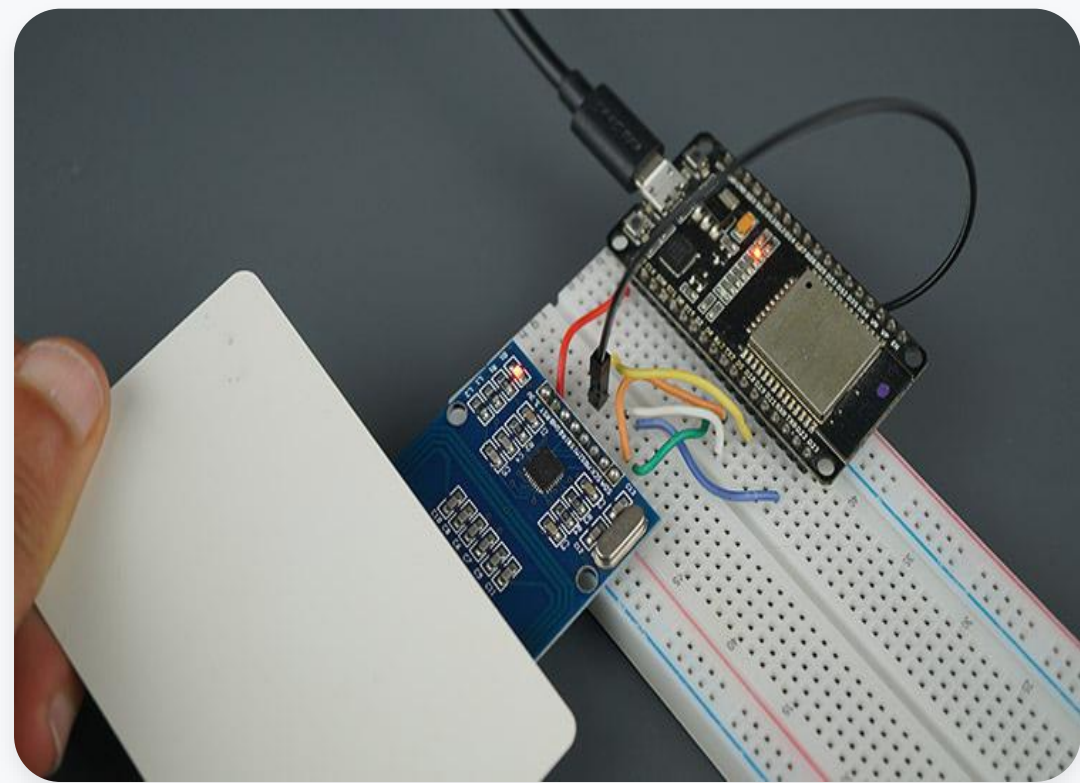
Contactless identification

Balanced & cost-effective

Mature technology

Scalable deployment

RFID TECHNOLOGY



RFID Frequency Bands

Band	Frequency	Range	Application
LF	125-134 kHz	Up to 10 cm	Access control
HF	13.56 MHz	Up to 1 meter	Our choice (RC522)
UHF	860-960 MHz	Up to 12 m	Supply chain
NFC	13.56 MHz	< 10 cm	Mobile payments

IoT in Education & Competitive Analysis

IoT APPLICATIONS IN EDUCATION



Smart Classrooms
Adaptive environments with sensors



Student Monitoring
Safety & location tracking



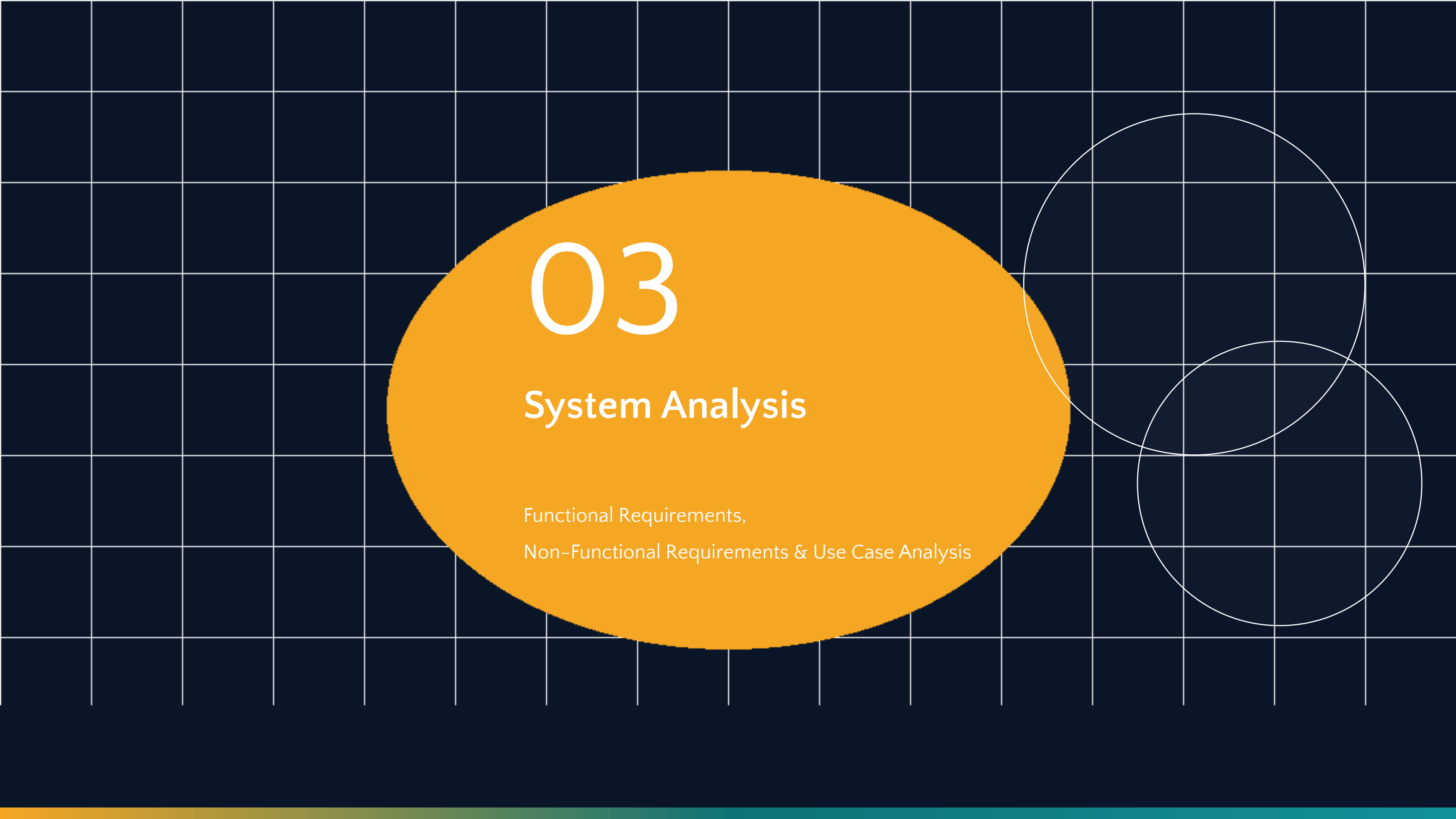
Data-Driven Decisions
Performance analytics & insights

COMPETITIVE COMPARISON

System	Offline	RFID	Mobile	RBAC	AI	Gradebook
Sharma et al.	X	✓	X	None	X	X
Khan et al.	Partial	✓	X	Basic	X	X
Patel & Gupta	X	✓	X	None	X	X
Al-Taee et al.	✓	X	✓	None	X	X
This Project	✓	✓	✓	6-Tier	✓	✓



Market Gap Identified: No existing solution combines offline-first architecture, multi-faculty RBAC, comprehensive gradebook integration, AI-powered analytics, and mobile application support in a single deployable package.



03

System Analysis

Functional Requirements,
Non-Functional Requirements & Use Case Analysis

Functional Requirements

Category	Key Requirements	Priority
User Management	Registration with role assignment (6 tiers), JWT authentication, bcrypt password hashing, account blacklisting, profile management	High
Academic Structure	Multi-faculty support (Engineering, Medicine, Business), department hierarchies, course definitions, session scheduling, student enrollment	High
Attendance	RFID card scanning via ESP32, one-scan-per-student uniqueness (409 Conflict), session validation, real-time monitoring, report generation	High
Gradebook	40/60 attendance-to-assessment weighting, multiple assessment types, weight validation (100%), grade finalization with commit/rollback	Medium
IoT Device Mgmt	ESP32 device registration, classroom mapping, RFID discovery mode, connectivity monitoring, Wi-Fi configuration management	Medium
Analytics	AI-powered academic insights (RAG), at-risk student reports, Google Sheets export, system health monitoring & telemetry	Low

System Actors & RBAC Model

ROLE-BASED ACCESS CONTROL

Role	Alias	Data Scope	Key Capabilities
Super Admin	super_admin	Global	Full system access, all faculties, user management, system configuration
Admin	admin	Faculty	Faculty CRUD, user approvals, IoT device management
Doctor	doctor	Department	Gradebook management, assessment oversight, grade finalization
Instructor	instructor	Course	Session management, quiz creation, IoT discovery mode
Engineer	engineer	Course	Session management, student scoring, attendance monitoring
Student	student	Self-only	View attendance records, grades, personal profile

PRIMARY USE CASES



UC-01: Record Attendance

Student taps RFID card on ESP32 reader.
System validates card, records timestamp, returns LED feedback.



UC-02: Manage Sessions

Instructor creates sessions, monitors real-time attendance, views statistics upon completion.



UC-03: Grade Reports

Doctor reviews assessments, verifies attendance calculations, commits final 40/60 grades.



UC-04: Configure System

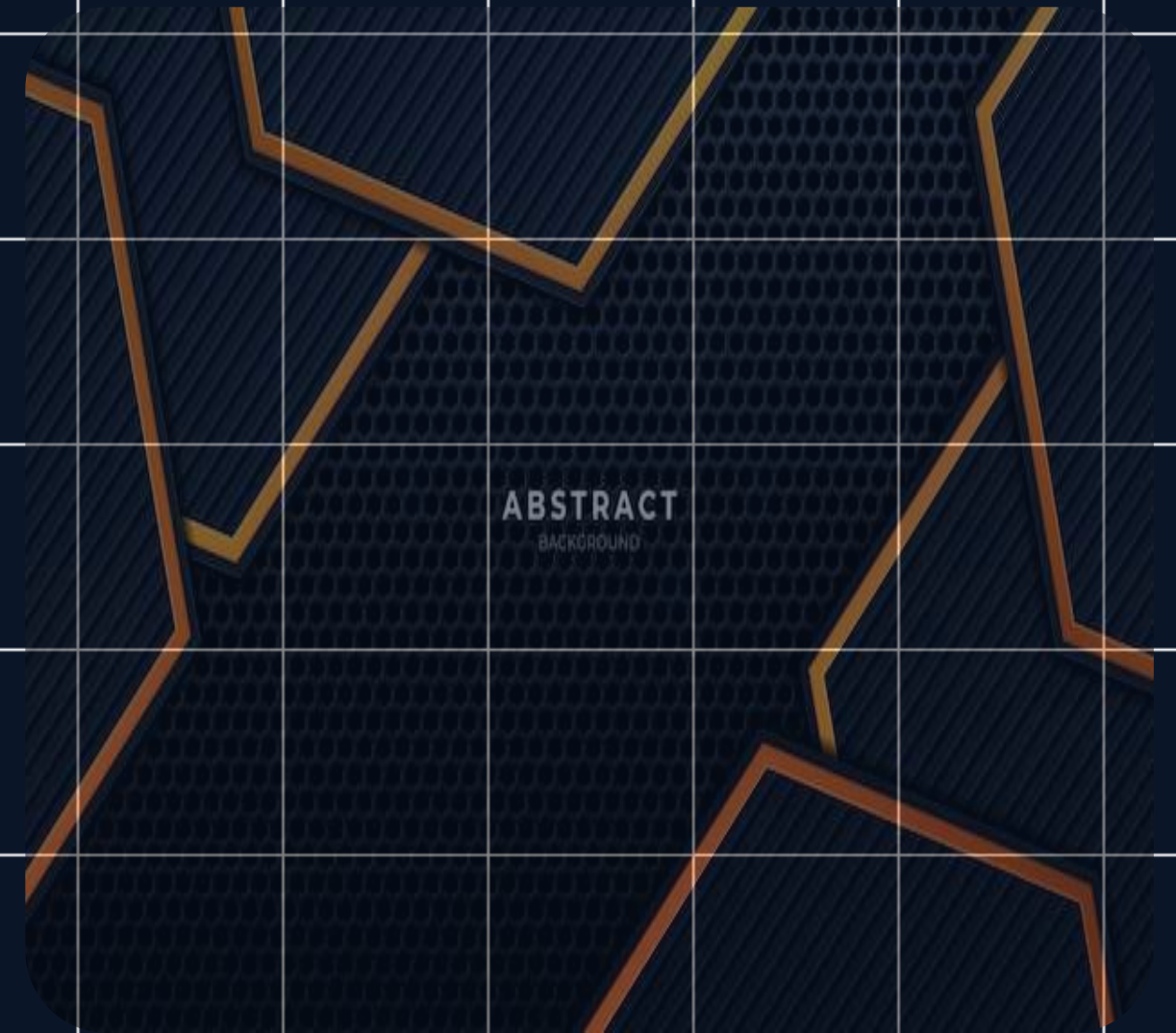
Super Admin configures university identity, structures, users, devices, and grading policies.

04

CHAPTER FOUR

System Design

Architecture, Database Design,
API Design & User Interface



Five-Layer System Architecture



Presentation Layer

React Web Dashboard + Kotlin Android App
RESTful API communication via JSON

React 18.3 | TailwindCSS | Recharts | Kotlin | Jetpack Compose



Application Layer

FastAPI business logic engine
16 route modules, RBAC, authentication

FastAPI 0.110 | Python 3.11+ | Pydantic v2 | JWT | bcrypt



Data Access Layer

SQLAlchemy 2.0 ORM with async support
23 entities, Alembic migrations

PostgreSQL 14 | SQLite | asyncpg | SQLAlchemy 2.0



Infrastructure Layer

Raspberry Pi 4, Nginx reverse proxy
Wi-Fi AP, PM2 process management

Raspberry Pi 4 | Nginx | hostapd | dnsmasq | PM2



IoT Device Layer

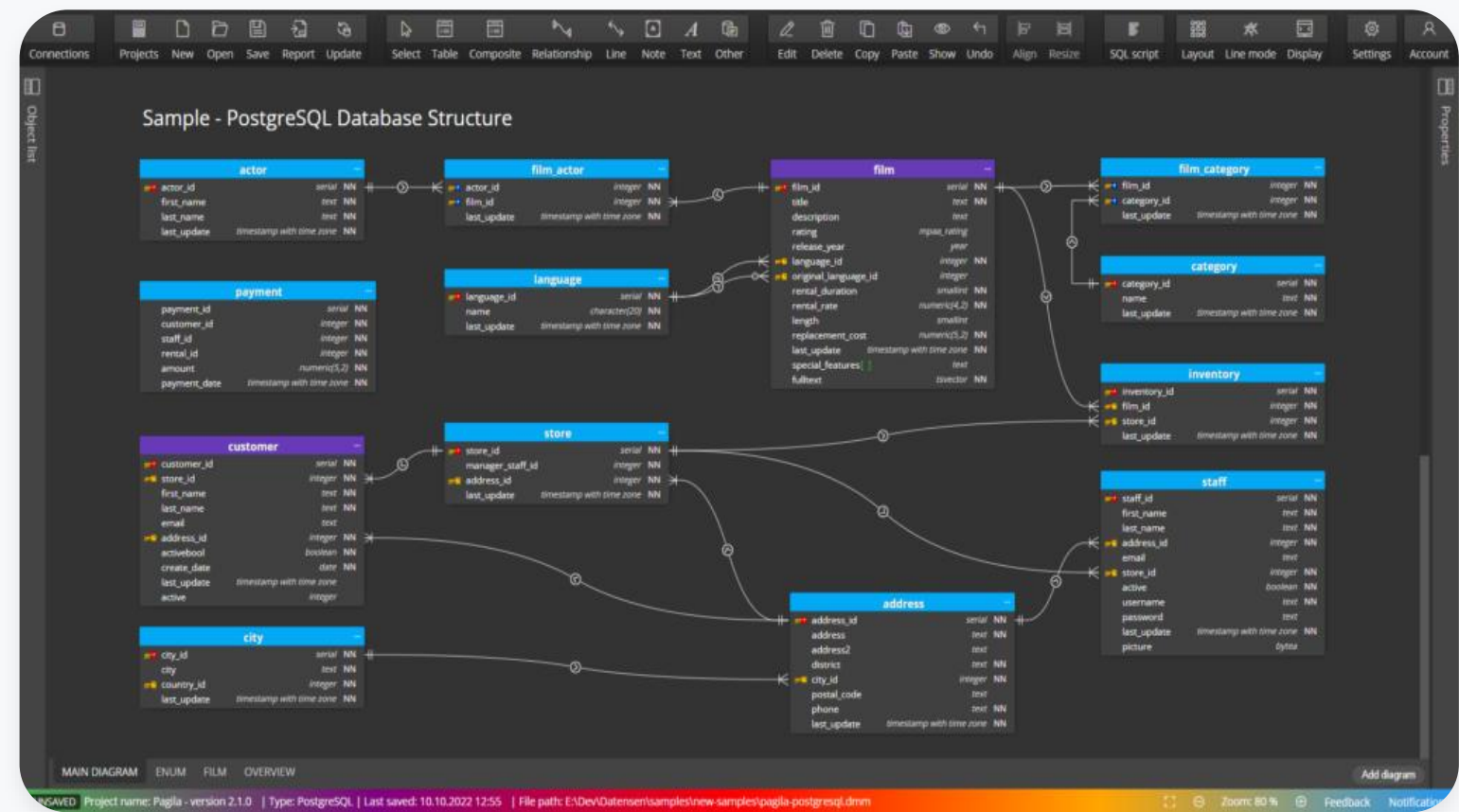
ESP32 microcontrollers + RC522 RFID readers
LED feedback, discovery mode, Wi-Fi

ESP32 DevKit | RC522 | Arduino | PlatformIO

Database Design & API Architecture

DATABASE ENTITIES (23 TABLES)

Entity	Description
User	System users with credentials & roles
Faculty	Top-level academic organizational unit
Department	Subdivision within a faculty
Course	Academic course with credit hours
Student	Student profile with enrollment info
Session	Scheduled class with time & location
Attendance	Individual attendance record per session
Assessment	Graded evaluation (quiz, exam, etc.)
GradeResult	Final computed grade per course
Device	Registered ESP32 with location mapping



API ENDPOINT GROUPS (16 MODULES)

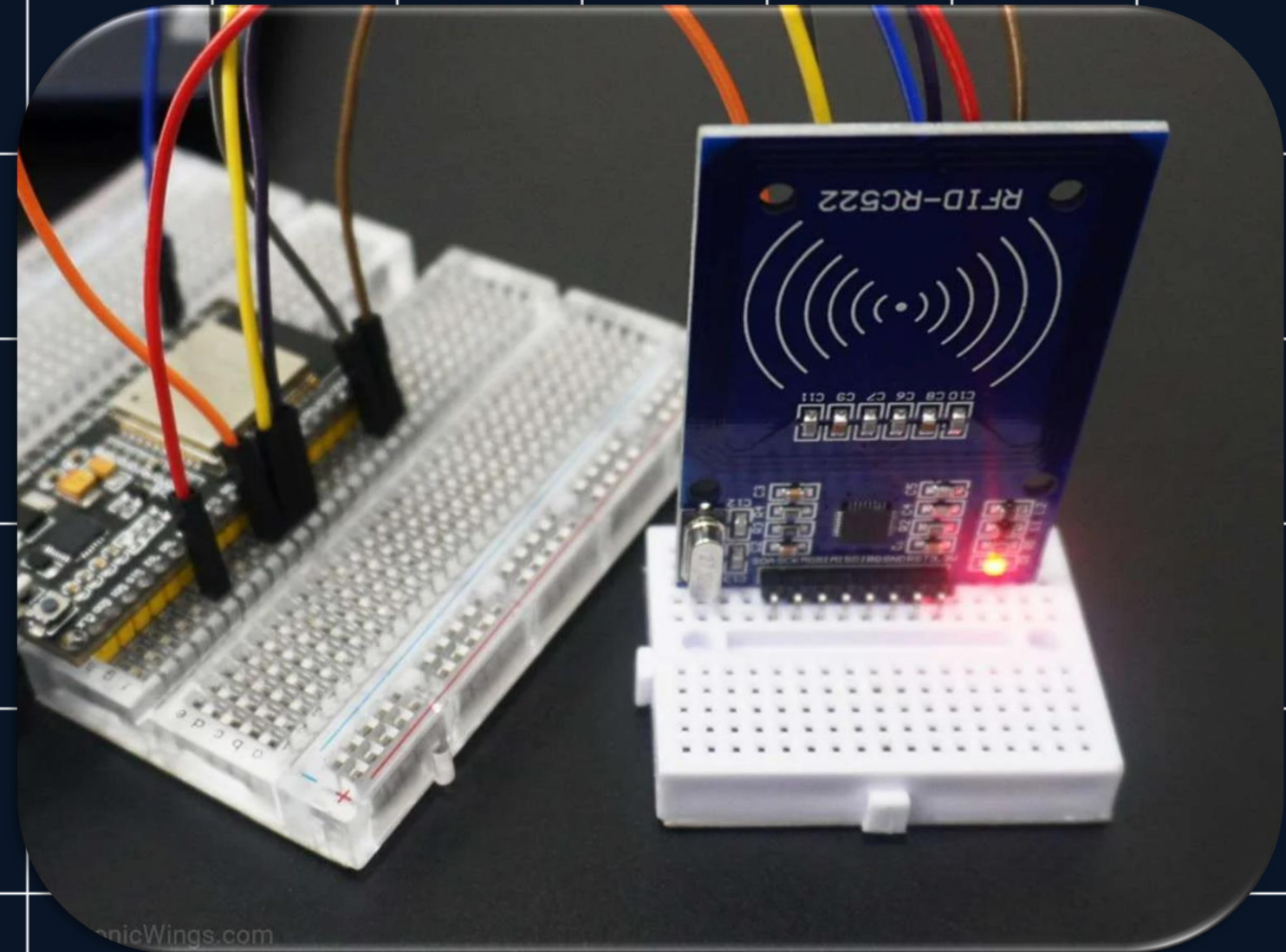
Module	Size	Description
admin	105 KB	System administration, user management, configuration
auth	-	Authentication, token management, password reset
sessions	-	Session CRUD, scheduling, activation
attendance	-	Attendance recording, validation, reporting
gradebook	-	Grade calculation, finalization, reports
monitoring	45 KB	System health, telemetry, device status
AI + IoT	-	RAG query engine, academic insights, device management

05

CHAPTER FIVE

Implementation

Development Environment, Backend,
Frontend, Mobile App & IoT Firmware



Technology Stack & Backend

TECHNOLOGY STACK

Component	Technology	Version / Details
Backend	FastAPI	0.110 / Python 3.11+
Frontend	React	18.3.1 / TailwindCSS
Mobile	Kotlin	Jetpack Compose / MVVM
IoT Firmware	ESP32	Arduino / PlatformIO
Database	PostgreSQL	14 / SQLite (dev)
Server	Raspberry Pi 4	PM2 / Nginx / hostapd

BACKEND IMPLEMENTATION

app/api/routes/ — 16 route modules

admin (105KB), monitoring (45KB), auth, sessions,
attendance, assessments, gradebook, AI, IoT

app/models/ — 23 SQLAlchemy ORM models

app/schemas/ — Pydantic v2 validation

app/services/ — 8 business logic modules

app/ai/ — RAG-style query engine

Security Implementation

- **Password Hashing** — bcrypt work factor 12
- **JWT Tokens** — 30min access / 7day refresh
- **RBAC Enforcement** — API-level role checks
- **Data Scoping** — Department-filtered queries
- **Blacklist** — 403 for suspended accounts
- **HS256** — Secure token signing

AI Integration (RAG)

Retrieval-Augmented Generation query engine processes natural language queries:

"Which students are at risk of failing?"

Retrieves relevant data from gradebook, attendance, and assessments to generate contextual academic insights and early warnings.

Web Dashboard, Mobile App & IoT Firmware



Web Dashboard

- 37 page components
- Recharts data visualization
- TailwindCSS styling (14KB)
- Role-based navigation
- Real-time monitoring
- Theme support

Pages:

Dashboard, Students, Courses, Gradebook, Sessions, Chatbot



Android Mobile App

- 21 feature packages
- MVVM architecture
- Jetpack Compose UI
- NFC card reading
- Offline data caching
- Material Design 3

Features:

Dashboard, Sessions, Attendance, Gradebook, Chatbot, Profile

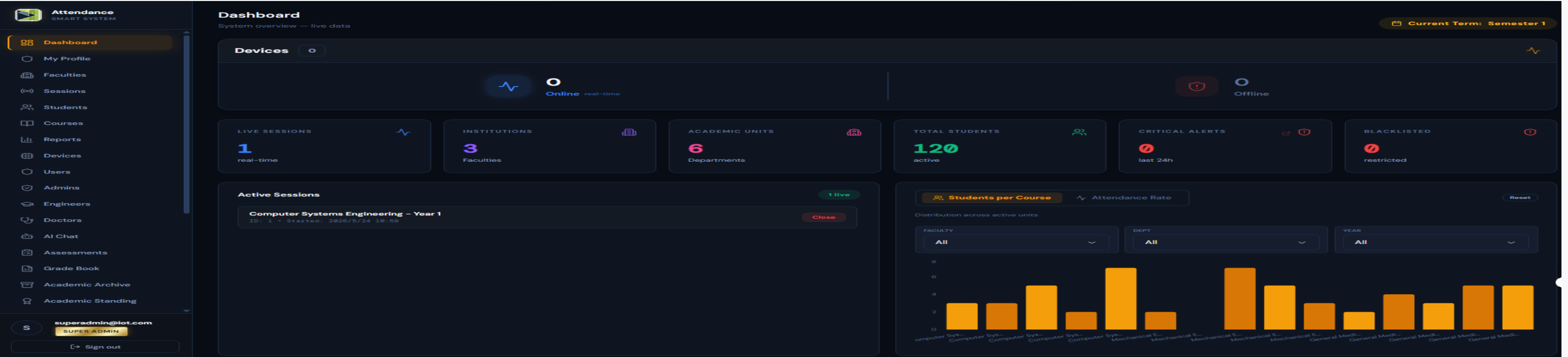


IoT Firmware (ESP32)

- ESP32 + RC522 module
- 13.56 MHz HF RFID
- Wi-Fi connectivity
- NTP time sync
- LED/buzzer feedback
- Discovery mode

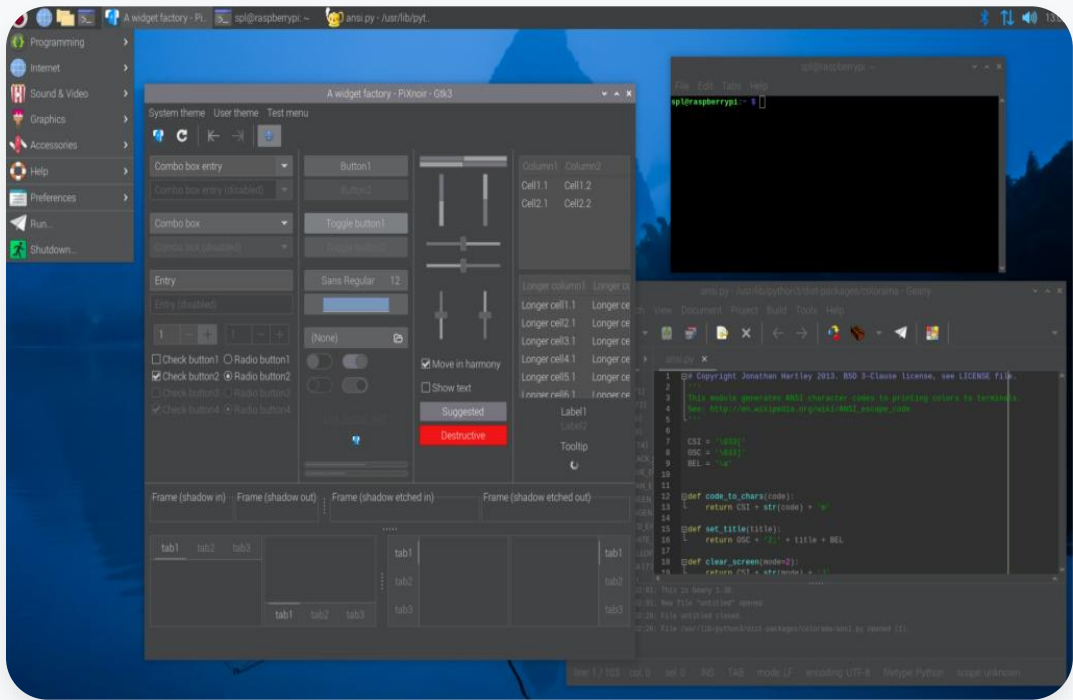
Operation:

Scan card, send UID, get response, LED feedback



Deployment & Infrastructure

RASPBERRY PI 4 DEPLOYMENT



NETWORK ARCHITECTURE

SSID: SmartAttendance (broadcast by Pi)

Domain: otu.university (dnsmasq)

IP: 192.168.1.7 (resolved internally)

Client Access Flow:

1. Connect to SmartAttendance Wi-Fi
2. Navigate to otu.university
3. Nginx routes to React or FastAPI

Complete Technology Stack on Pi 4

- **OS:** Raspberry Pi OS (Debian-based) headless
- **Process Management:** PM2 auto-restart on failure
- **Reverse Proxy:** Nginx serves React + proxies API
- **Database:** PostgreSQL 14 optimized for ARM
- **Wi-Fi AP:** hostapd (SmartAttendance SSID)
- **DNS/DHCP:** dnsmasq (otu.university domain)

Configuration Management

- **ecosystem.config.js** — PM2 processes
- **.env** — DB URL, JWT secret, API config
- **nginx.conf** — Reverse proxy rules
- **platformio.ini** — ESP32 build config

Startup wizard for new deployments

06

CHAPTER SIX

Testing & Results

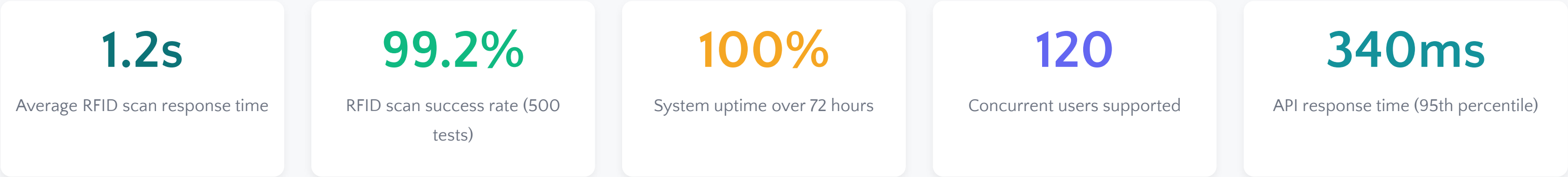
Testing Methodology, Unit Testing,
Integration Testing & System Results

Testing Results & Performance

TEST ENVIRONMENT

Hardware: Raspberry Pi 4 (4GB) | **Network:** 10 client devices | **Test Data:** 3 faculties, 8 departments, 15 courses, 120 students | **IoT:** 3 ESP32 units

KEY PERFORMANCE METRICS



DETAILED TEST RESULTS

Test Case	Result	Notes
RFID scan — valid card	PASS	1.2s avg response time
RFID scan — duplicate	PASS	409 Conflict returned correctly
RFID scan — unenrolled	PASS	403 Forbidden returned correctly
User login — valid	PASS	JWT token issued with correct payload
User login — invalid	PASS	401 Unauthorized returned
Grade calculation	PASS	40/60 weighting verified correct
RBAC enforcement	PASS	All 6 roles verified
72-hour uptime test	PASS	100% availability, zero data corruption

07

CHAPTER SEVEN

Conclusion

Summary, Key Achievements
& Future Work

Achievements & Future Work

KEY ACHIEVEMENTS



Offline-First System

Full operation without internet, designed for Iraqi universities



6-Tier RBAC

Hierarchical role system from Super Admin to Student



Multi-Faculty Support

Engineering, Medicine, Business with 100+ students



AI Academic Insights

RAG-powered queries for at-risk student identification



Sub-2s RFID Response

1.2s average scan time with 99.2% success rate



Pi 4 Deployment

Self-contained portable system with startup wizard

FUTURE WORK



Biometric Integration

Add fingerprint and facial recognition as secondary authentication methods



iOS Application

Develop native iOS app using SwiftUI for Apple device users



Cloud Sync

Enable multi-campus data synchronization with cloud backup



Predictive Analytics

ML-based early warning system for student performance prediction

THANK YOU

QUESTIONS & DISCUSSION

Smart IoT Attendance System Ecosystem

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ACADEMIC YEAR 2025-2026